

HumanEd: "Oar" Robot

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1 Overview



Figure 1: Mockup of Oar

The "Oar" robot is a humanoid robot comprised of a rigidly-supported torso, a head placed upon an actuated neck, and two 7 degree of freedom robotic arms. A basic mockup of Oar is shown in Figure 1.

The design goals for Oar are as follows:

- Demonstrate the strengths, capabilities, and pitfalls of 3D printed actuator gearhead designs and serve as a design study that may inform future HumanEd humanoid robots.
- Provide a reliable humanoid arm system capable of accurately positioning the end-effector for experimental and demonstratory purposes. Experimental usages includes testing of end-effector designs (grippers, hands), a real-world interface for computer vision experiments, and a follower-arm for teleoperations experiments. Demonstrations may include various tasks that could be performed at fairs or open days at which the society of HumanEd may be present.
- Provide a reasonable suite of sensors that can be interfaced with for experiments. Such experiments may include computer vision or reinforcement learning projects.

This proposal will focus primarily on the development of the robot's arms as they are considered to be the primary engineering challenge, however some notes on the body, neck/head, and end-effector will still be made.

1.1 A note on the name

The name "Oar" has been derived from the first line of the poem "Basking Shark" by Scottish poet Norman MacCaig, which reads:

*To stub an **oar** on a rock where none should be*

The poem is a discussion on themes of the origins of life and the place of humanity in the timeline of evolution. The decision to choose this poem to name the robot reflects the robot's intended position as HumanEd's first full-scale humanoid robot. It is also intended for an inscription of the poem to be placed somewhere on the robot in commemoration of this poem.

2 Arm Design

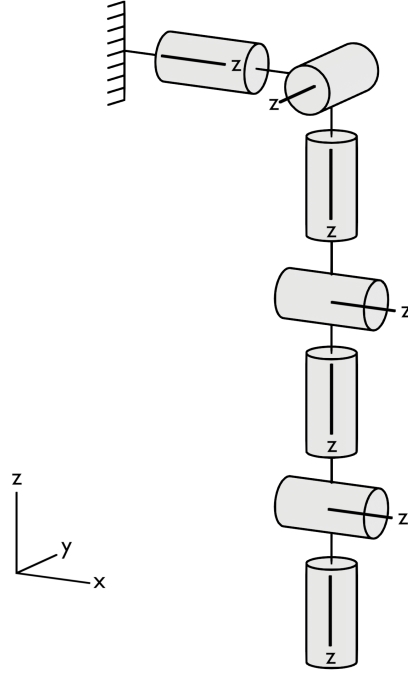


Figure 2: Joint diagram of Oar's left arm

The arms of Oar will follow the common humanoid arm model of a 3-DOF shoulder and a 3-DOF wrist connected by a 1-DOF elbow. Figure 2 shows the joint diagram of the left arm.

The arm is /// talk about length ///

2.1 Forward Kinematics & Preliminary Torque Analysis

All of the joints shown in Figure 2 are revolute joints. For the purpose of the forward kinematics analysis, all three of the shoulder joints will be considered coincident, and all three of the wrist joints will also be considered coincident. The vertical distance between the shoulder joints and the elbow joint is taken as L_1 , and the distance between the elbow joint and the wrist joints as L_2 .

The resting position of the end-effector frame $\{b\}$ is a right-handed set of axes rotated by π radians about the x-axis such that the z_b axis points downwards.

The forward kinematics of the arm can then be defined via the method as described by Lynch and Park [1, § 4.1]. $\theta \in \mathbb{R}^7$ is the vector of all joint positions of the arm, and $M \in SE(3)$ is the transformation matrix of the end-effector when the arm is in its resting position, defined as

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & -(L_1 + L_2) \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3)$$

The screw axes $\mathcal{S}_i$, $i = 1, \dots, 7$, represented in the space frame, can be seen as:

$$\begin{aligned} \mathcal{S}_1 &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{S}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{S}_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{S}_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -L_1 \\ 0 \end{bmatrix} \\ \mathcal{S}_5 &= \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}, \quad \mathcal{S}_6 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}, \quad \mathcal{S}_7 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix} \end{aligned}$$

Which yields the forward kinematics equation where $T(\theta) \in SE(3)$ is the transformation matrix describing the position and orientation of the end-effector given the joint positions θ :

$$T(\theta) = {}_e[\mathcal{S}_1]\theta_1 {}_e[\mathcal{S}_2]\theta_2 {}_e[\mathcal{S}_3]\theta_3 {}_e[\mathcal{S}_4]\theta_4 {}_e[\mathcal{S}_5]\theta_5 {}_e[\mathcal{S}_6]\theta_6 {}_e[\mathcal{S}_7]\theta_7 M \quad (1)$$

Equation 1 can then be used to find the space Jacobian $J_s(\theta) \in \mathbb{R}^{6 \times 6}$ of the arm's end-effector via the expression:

$$J_{si}(\theta) = \left[\text{Ad}_{{}_e[\mathcal{S}_1]\theta_1 \dots {}_e[\mathcal{S}_{i-1}]\theta_{i-1}} [{}_e[\mathcal{S}_i]\theta_i] \right] \mathcal{S}_i \quad (2)$$

where $J_{si}(\theta)$ is the i th columns of $J_s(\theta)$. This can then be used to perform a statics analysis of the arm via the equation

$$\tau = J_s^T(\theta) \mathcal{F}_s \quad (3)$$

where $\mathcal{F}_s$ is a wrench applied to the end-effector, and τ is the resulting vector of torques required in each joint to resist $\mathcal{F}_s$.

The method for performing this analysis is detailed in [1, § 5] and for this robot results in matrices that are too large and complex to be reasonably computed by hand for every possible configuration of interest. As such Equations, 1, 2, and 3 were implemented into a python script to complete the analysis.

Using the python script, a variety of configurations were tested for a 600 g payload.

The tested configurations showed that the maximum torque required of the arm's joints would take place in the shoulder pitch, shoulder roll, and elbow, and would be roughly of a magnitude of 4 Nm.

This result was then used to inform the design of various prototype gearheads.

The prototypes are detailed in § 3.

2.2 Gearhead Efficiency Requirement Analysis

3 Gearhead Design

Cycloidal drives have been chosen for the actuators of the arm. This is primarily due to their affinity for being produced using FDM printing when compared to planetary or strain-wave gear systems, wherein the former presents difficult to achieve tolerances (as well as comparatively low reduction ratios) and the latter requires the development of a flexible element.

The following sections detail the development of a prototype gearhead. The model of stepper motor used was the 17HS3401S, which was ran at a speed of 100 RPM and measured to have a dynamic torque of 0.17 Nm ¹.

The output torque produced by each prototype gearhead was tested using various masses suspended from a 30 cm bar of $20 \text{ mm} \times 40 \text{ mm}$ aluminium extrusion attached from the centerpoint of the gearhead’s output.

Table 1 shows a summary of the results for all three prototypes.

No.	Reduction Ratio	Mass (g)	Maximum Measured Output Torque (Nm)	Efficiency
1	24:1	420	1.0	25%
2	48:1	460	1.4	17%
3	48:1	740	3.1	38%

Table 1: Summary of prototype gearheads.

3.1 First Prototype - 24:1

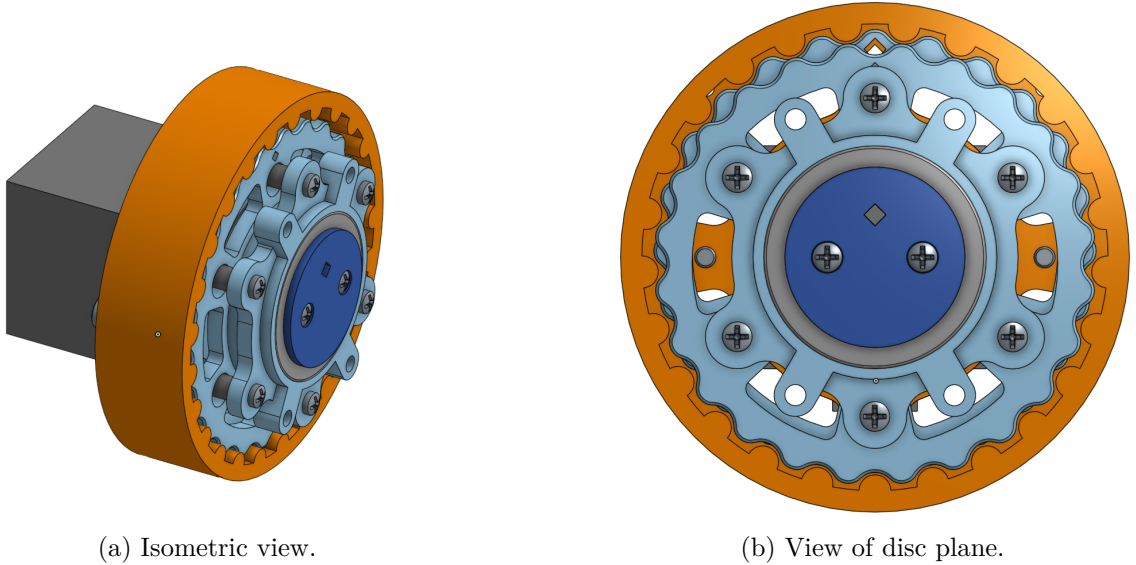


Figure 3: The prototype 24:1 gearhead design.

Figure 3 shows the design of the first prototype gearhead that was built and tested. This prototype was built with a 24:1 reduction ratio, half of the primary 48:1 reduction ratio calculated in § 2.1. Figure 3a shows an isometric view of the design with all housings included, Figure 3b shows a view from the plane of the gearhead’s cycloidal discs, with the output housing hidden for clarity.

The design is almost entirely comprised of FDM printed parts, with the only exceptions being the bearings, screws, and of course stepper motor.

¹Used to drive a rudimentary pulley of diameter 30 mm and found to be capable of lifting a maximum of 575 g.

The gearhead was seen to be able to achieve a maximum output torque of roughly 1.0 Nm, which gives an efficiency of about 25%. This is, needless to say, a poor result, however it did show that the design was able to successfully increase the output torque of the motor.

3.2 Second Prototype - 48:1

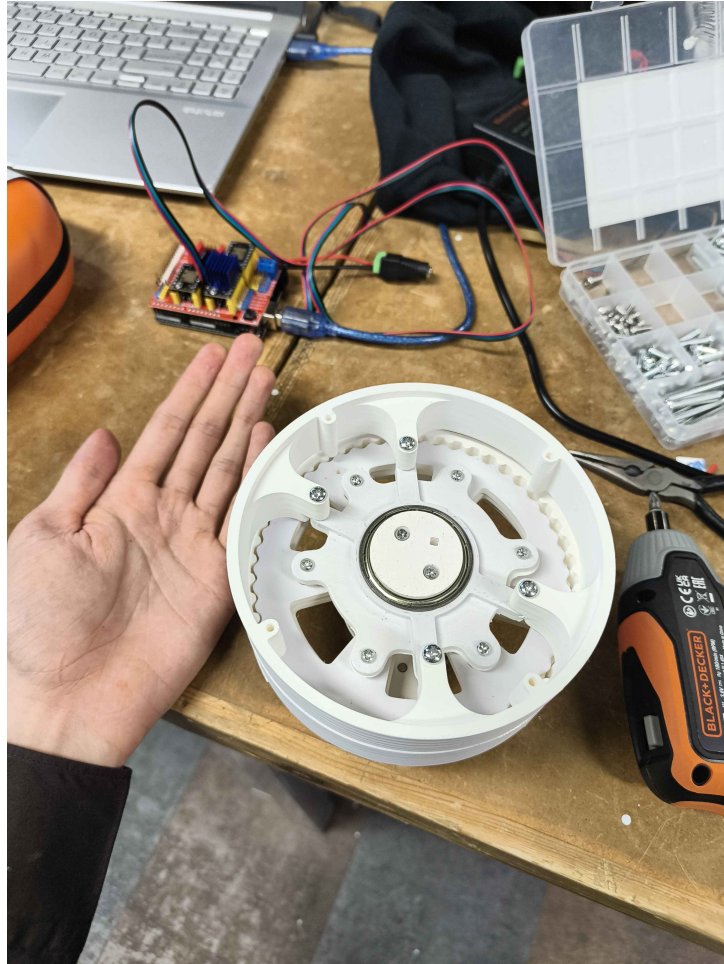


Figure 4: A photo of the second constructed prototype gearhead, author's hand for scale.

The primary cause of the low efficiency seen with the first prototype in § 3.1 was believed to be excess friction caused by poor tolerances and printed part surface finishes. The second prototype, shown in Figure 4, was produced using a higher quality printer in order to improve these tolerances and surface finishes. Due to added material availability, it was also decided to scale the prototype up to the full 48:1 design. The design of this second prototype was fundamentally the same as the first, only scaled up.

Despite the improved print quality, efficiency actually decreased from the initial prototype. The highest output torque achieved was roughly 1.4 Nm, which gives an efficiency of 17%.

3.3 Third Prototype - 48:1 with Steel Dowels

Following the low efficiencies of the first and second prototypes, as seen in § 3.1 and § 3.2 respectively, it became clear that the frictional forces present between the rough surfaces of FDM printed parts were too high to allow an effective gearhead to be produced without the usage of some lower frictional elements.

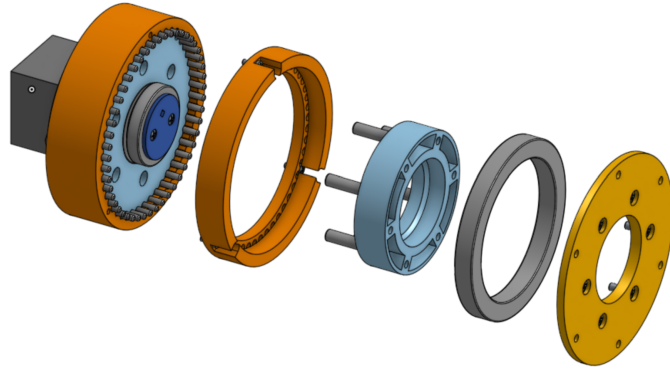


Figure 5: A partially exploded view of the third prototype gearhead.

As such, the third prototype gearhead, as seen in Figure 5, made use of steel dowels for both the inner and outer pins of the cycloidal drive. An additional large diameter bearing was also added to stabilise the output disc and prevent friction from its lateral warping, which was observed in the prior prototypes. PTFE synthetic grease was also applied as a lubricant to all surfaces subject to high friction.

This prototype was seen to achieve a maximum output torque of 3.1 Nm, giving an efficiency of 38%.

4 Body, Head, & Sensors

5 Points for Development & Timeline

References

- [1] K. M. Lynch and F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017, ISBN: 978-1-107-15630-2.